

Question 1

Each candidate must be provided with the following apparatus and materials:

- 1 25 cm³ of reconstituted milk solution with calcium, labelled **M + C**
This is prepared by adding 10 g of Coffee-mate[®] Original (do not use fat-free Coffee-mate[®]), to 100 cm³ hot, distilled water (approximately 80°C) and stirring. To this solution, add 0.05 g calcium chloride (CaCl₂·2H₂O – not anhydrous) and stir. This is sufficient for four candidates.

The milk solution must be prepared on the day of the examination and provided to the candidates at room temperature.

The solution should be provided to candidates in 100 cm³ beakers, labelled **M + C**.

- 2 25 cm³ of reconstituted milk solution without calcium, labelled **M**
This is prepared by adding 10 g of Coffee-mate[®] Original (do not use fat-free Coffee-mate[®]), to 100 cm³ hot, distilled water (approximately 80°C) and stirring. This is sufficient for four candidates.

The milk solution must be prepared on the day of the examination and provided to the candidates at room temperature.

The solution should be provided to candidates in 100 cm³ beakers, labelled **M**.

- [H] 3 20 cm³ of 1.5% lipase solution, labelled **E10**
This is prepared by adding 1.5 g of lipase powder to 70 cm³ of distilled water in a beaker while stirring. Make up to 100 cm³ with distilled water. This is sufficient for five candidates.

This must be prepared within **one hour** of the start of **Question 1** and be at room temperature for use by candidates.

The solution should be provided to candidates in 100 cm³ beakers, labelled **E10**. Candidates will be informed that **E10** is a 10% lipase solution.

- [H] 4 20 cm³ of 1.2% lipase solution, labelled **E5**
This is prepared by adding 1.2 g of lipase powder to 70 cm³ of distilled water in a beaker while stirring. Make up to 100 cm³ with distilled water. This is sufficient for five candidates.

This must be prepared within **one hour** of the start of **Question 1** and be at room temperature for use by candidates.

The solution should be provided to candidates in 100 cm³ beakers, labelled **E5**. Candidates will be informed that **E5** is a 5% lipase solution.

Before the examination, the activity of the lipase solution should be tested at about 40°C. Mix 2 cm³ of solution **M + C** with 2 cm³ of solution **T** and 3 cm³ of solution **A**. Add 2 cm³ of solution **E5**. The indicator should change colour from blue to yellow within 5 minutes. If there is no colour change within 5 minutes then use 2% lipase for solution **E5** and 2.5% lipase for solution **E10**. Do not inform candidates of the change in concentration or change the labelling of the solutions.

- 5 30 cm³ of sodium carbonate solution, labelled **A**
This is prepared by dissolving 0.30 g of anhydrous sodium carbonate in 80 cm³ of distilled water in a beaker and stirring to dissolve. Make up to 100 cm³ with distilled water. This is sufficient for three candidates.

The solution should be provided to candidates in 100 cm³ beakers, labelled **A**.

- 6 25 cm³ of Thymol blue solution, labelled **T**
This is prepared by dissolving 0.1 g of Thymol blue powder in 50 cm³ of 70% ethanol and making up to 250 cm³ with distilled water. This is sufficient for ten candidates.
- The solution should be provided to candidates in 100 cm³ beakers, labelled **T**.
- 7 four boiling tubes
- 8 two boiling tube racks to hold four boiling tubes
- 9 1 x 5 cm³ syringe and 3 x 3 cm³ syringes
- If there is a shortage of syringes, candidates can be instructed to re-use them as long as the syringes are thoroughly washed with distilled water before re-use.
- 10 stopwatch
- 11 glass marker pen
- 12 thermometer (-10°C to 110°C)
- 13 500 cm³ beaker labelled **water bath**, to act as a water-bath to contain four boiling tubes at a time
- 14 500 cm³ glass beaker, with water between 40°C and 45°C, labelled **hot water**
- The supervisor may use a thermostatically-controlled water-bath to provide additional hot water if requested by candidates.
- 15 access to a tap and sink
- 16 paper towels
- 17 a pair of goggles or eye protection
- 18 a pair of disposable gloves

Question 2

Each candidate must be provided with the following apparatus and materials:

- (i) Microscope with low and high power objectives e.g. X4 and X 40
- (ii) Slices (2mm thick) of potato sufficient for the student to cut at least 12 strips 80 mm long – Approximately 5 slices in a sealed Ziploc bag (they must not dry up).
The potato slicer is to be used to slice the potato.
- (iii) Part of a red fleshy scale leaf from an onion (at least 2 cm width). The material must be provided fresh in a sealed Ziploc bag.
- (iv) 50 cm³ of each of **0.1 M**, **0.5 M** and **1.0 M** sucrose solutions, labelled as such.
- [O] (v) 5 cm³ of 1mol cm⁻³ potassium nitrate solution in a container labelled **X**.
- [H] (vi) Dropper(3ml) for solution **X**
- (vii) 5 cm³ of **methylene blue** labelled as such. Prepared by dissolving 2 g in 100 cm³ of water. Skin Irritant/ Harmful.
- (viii) Filter paper (to substitute tissue paper for Q2 (a))
- (ix) Three 10 cm³ syringes
- (x) Marker for marking glassware
- (xi) Three very fine teat droppers to be used for Q2 (b)
- (xii) Ruler mm
- (xiii) Scalpel
- (xiv) Fine forceps
- (xv) Microscope slides and cover slips
- (xvi) Mounted needle
- (xvii) Three tiles
- (xviii) Seven test tubes (15 x 125 mm approx); test tube rack.
- (xix) Beaker (250 cm³) containing 100 cm³ distilled water, marked **Water for washing**
- (xx) Paper towels
- (xxi) Petri dish